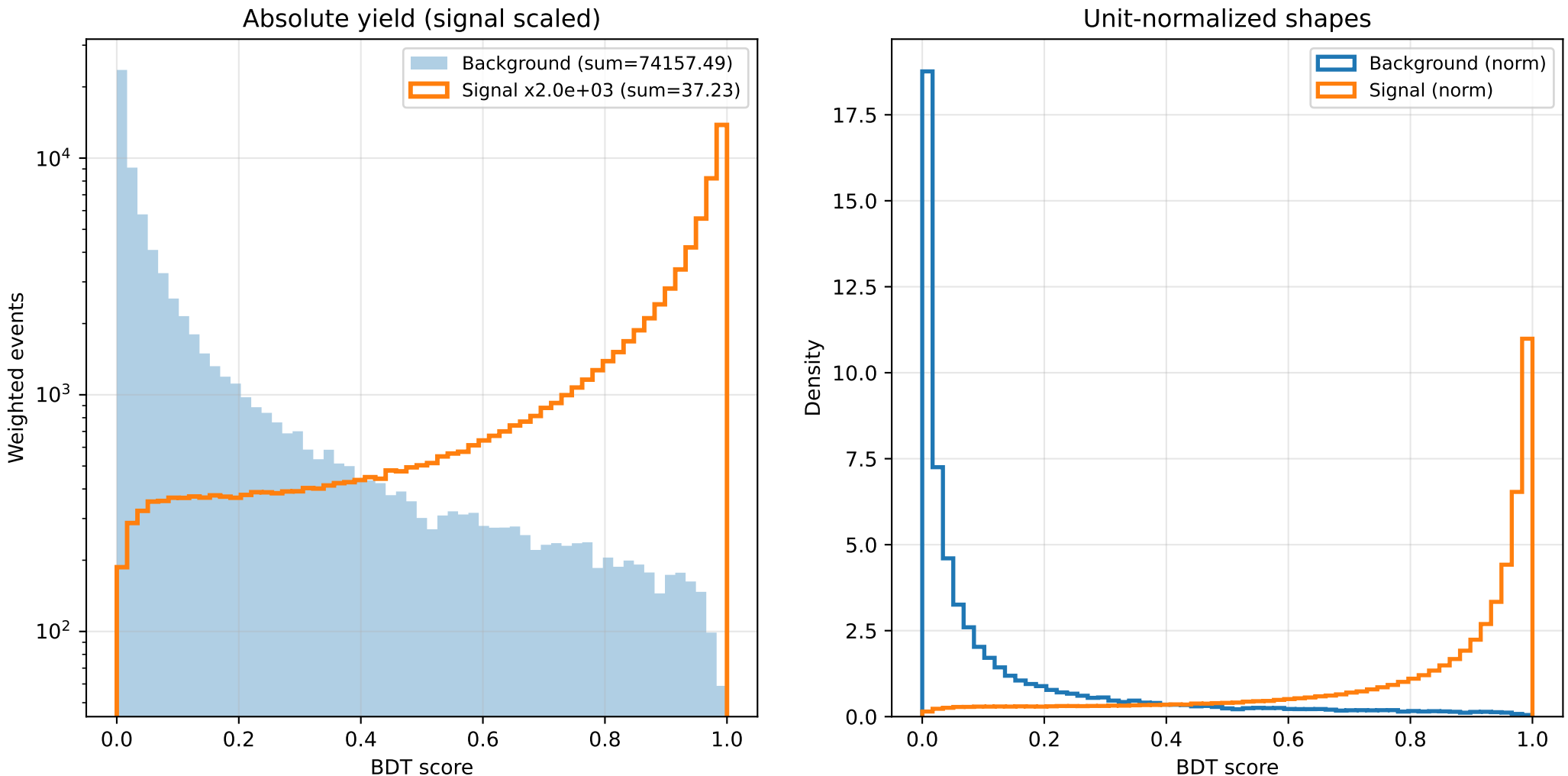
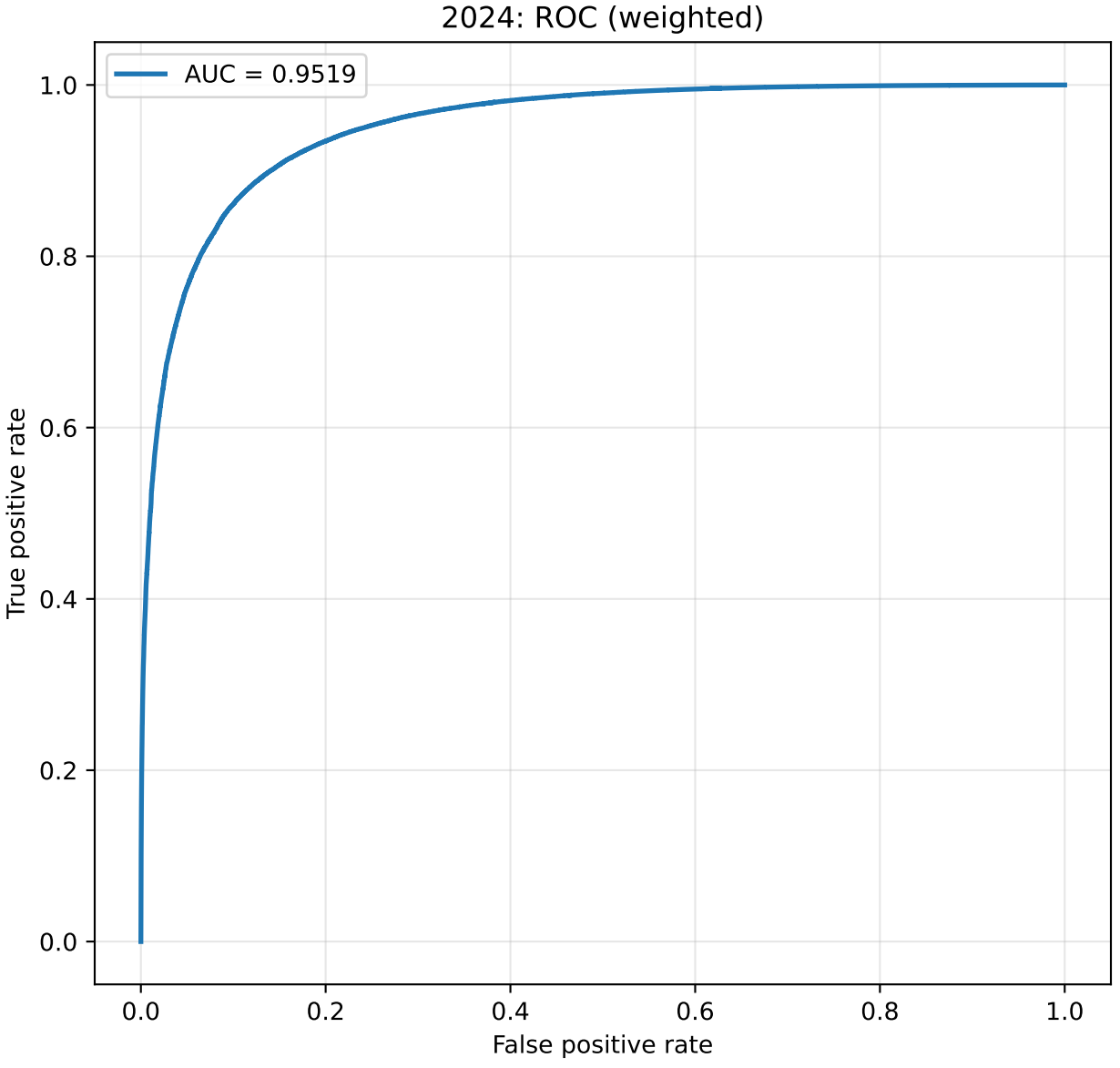
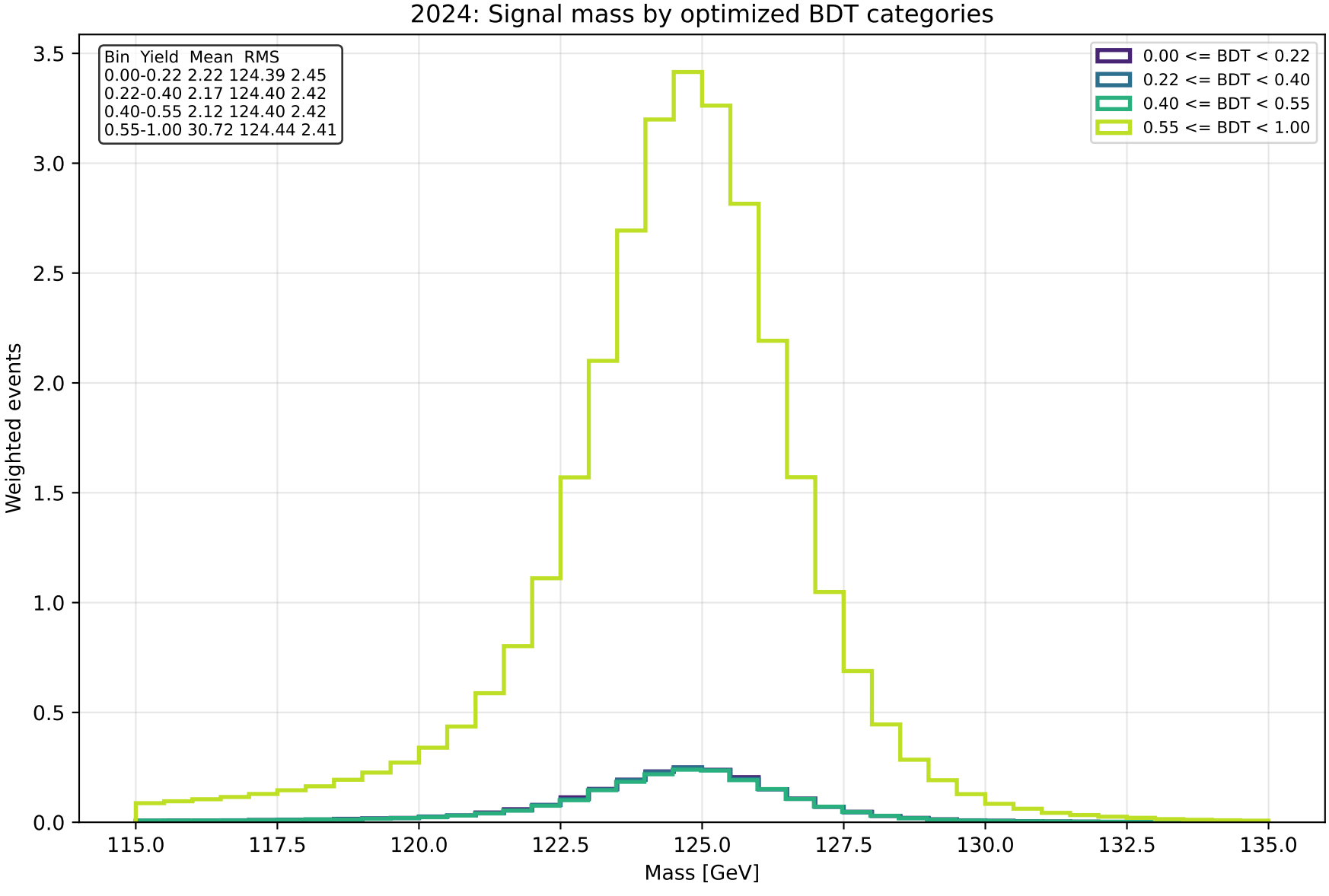
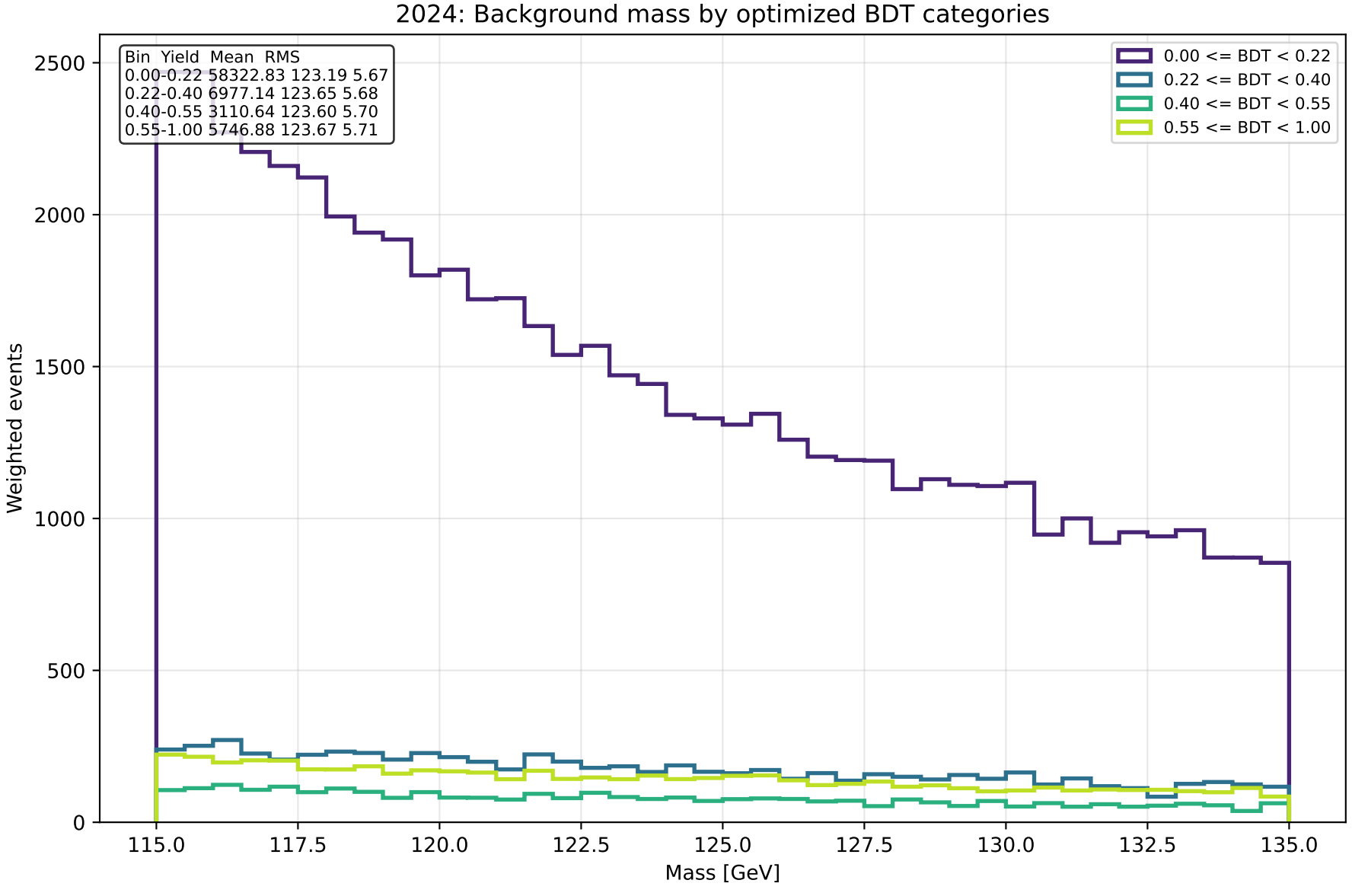

2024: BDT scores in 115-135 GeV
Absolute yield (signal scaled)
Background (sum=74157.49)
Signal x2.0e+03 (sum=37.23)
Weighted events
10^4
10^3
10^2
0.0
0.2
0.4
0.6
0.8
1.0
BDT score
Unit-normalized shapes
Background (norm)
Signal (norm)
Density
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0.0
0.2
0.4
0.6
0.8
1.0
BDT score

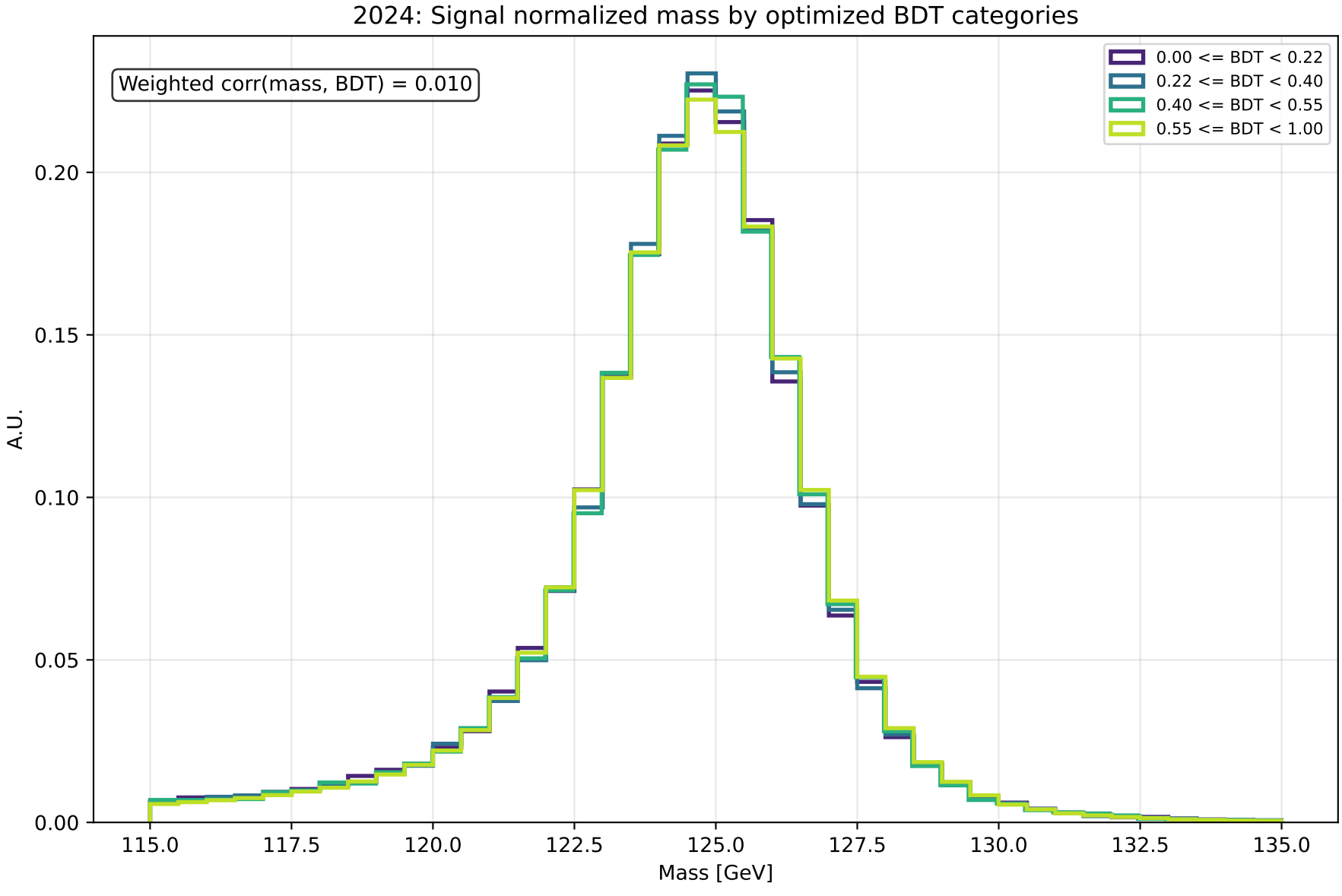

2024: ROC (weighted)
AUC = 0.9519
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0

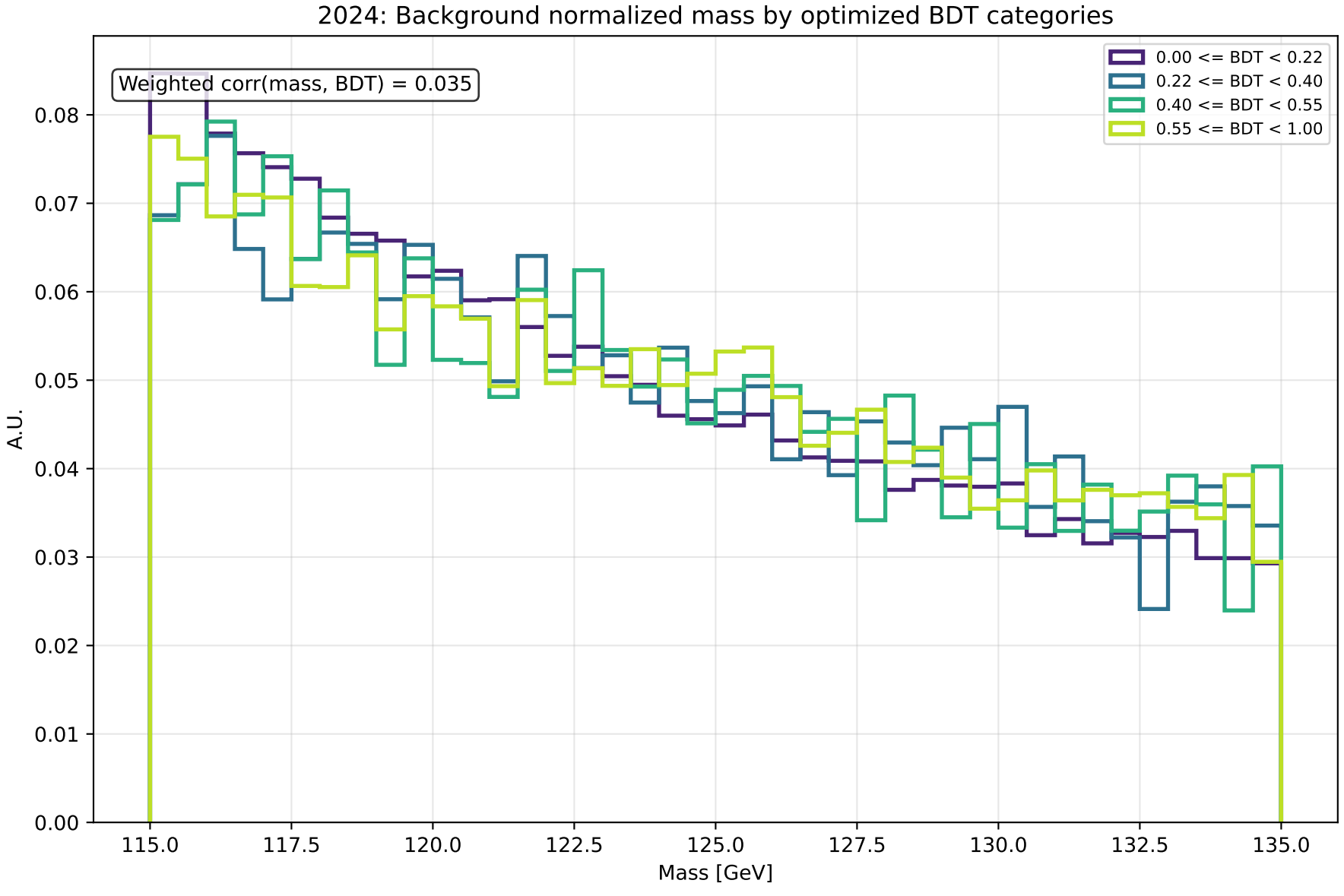
2024: Multi-bin BDT categorization (no hard cut)
Optimized N=4, Zexp(profile)=2.958, monotonic S/B=True

Bin	BDT range	S	B	S/B	S/sqrt(B)
1	0.000-0.218	2.219	58322.830	3.81e-05	0.009
2	0.218-0.400	2.168	6977.136	0.000311	0.026
3	0.400-0.546	2.115	3110.638	0.00068	0.038
4	0.546-1.000	30.723	5746.883	0.00535	0.405


2024: Signal mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.22 2.22 124.39 2.45
0.22-0.40 2.17 124.40 2.42
0.40-0.55 2.12 124.40 2.42
0.55-1.00 30.72 124.44 2.41
0.00 <= BDT < 0.22
0.22 <= BDT < 0.40
0.40 <= BDT < 0.55
0.55 <= BDT < 1.00
Weighted events
Mass [GeV]
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0


2024: Background mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.22 58322.83 123.19 5.67
0.22-0.40 6977.14 123.65 5.68
0.40-0.55 3110.64 123.60 5.70
0.55-1.00 5746.88 123.67 5.71
0.00 <= BDT < 0.22
0.22 <= BDT < 0.40
0.40 <= BDT < 0.55
0.55 <= BDT < 1.00
Weighted events
0
500
1000
1500
2000
2500
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]


2024: Signal normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.010
0.00 <= BDT < 0.22
0.22 <= BDT < 0.40
0.40 <= BDT < 0.55
0.55 <= BDT < 1.00
A.U.
0.00
0.05
0.10
0.15
0.20
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

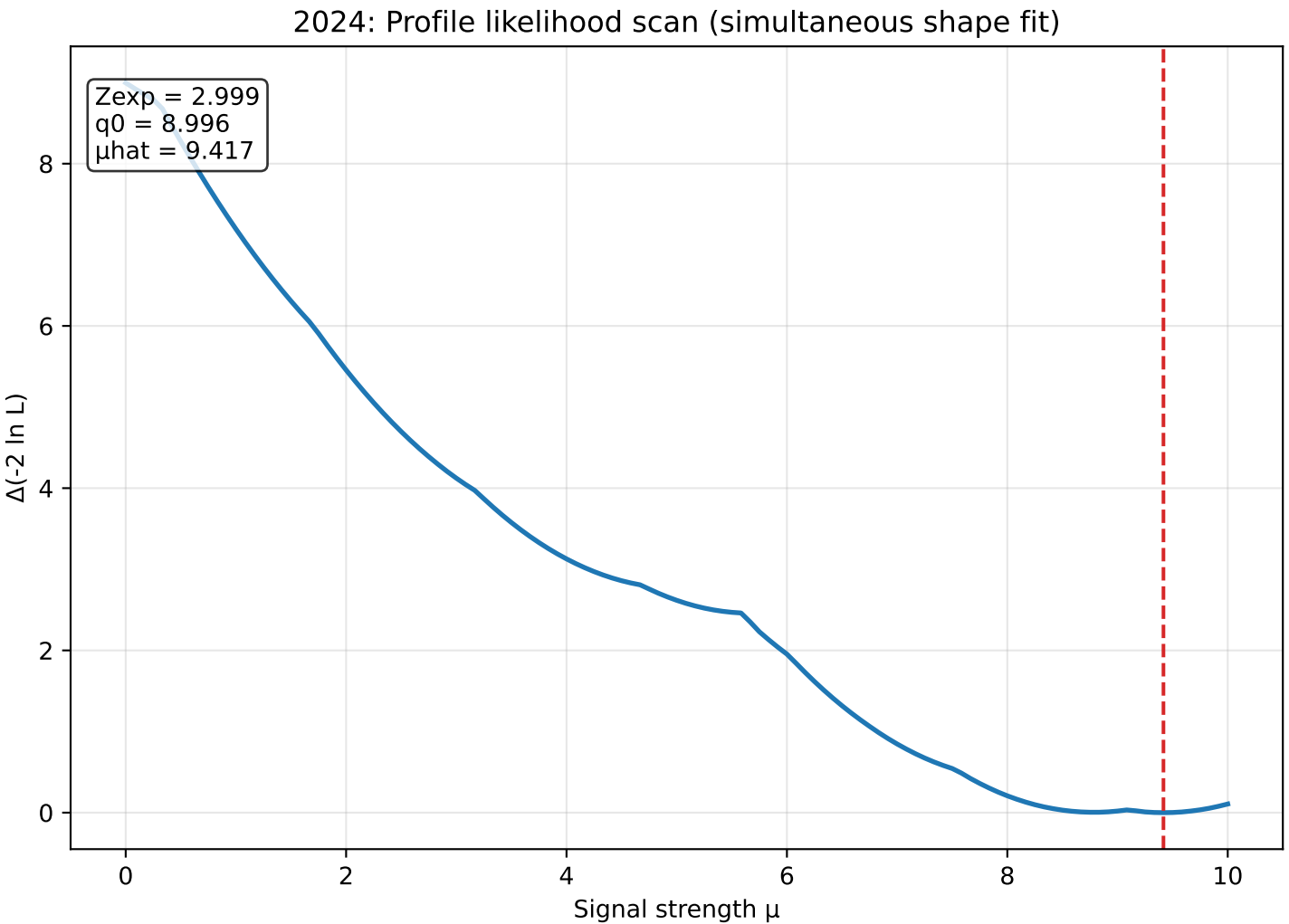

2024: Background normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.035
0.00 <= BDT < 0.22
0.22 <= BDT < 0.40
0.40 <= BDT < 0.55
0.55 <= BDT < 1.00
A.U.
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

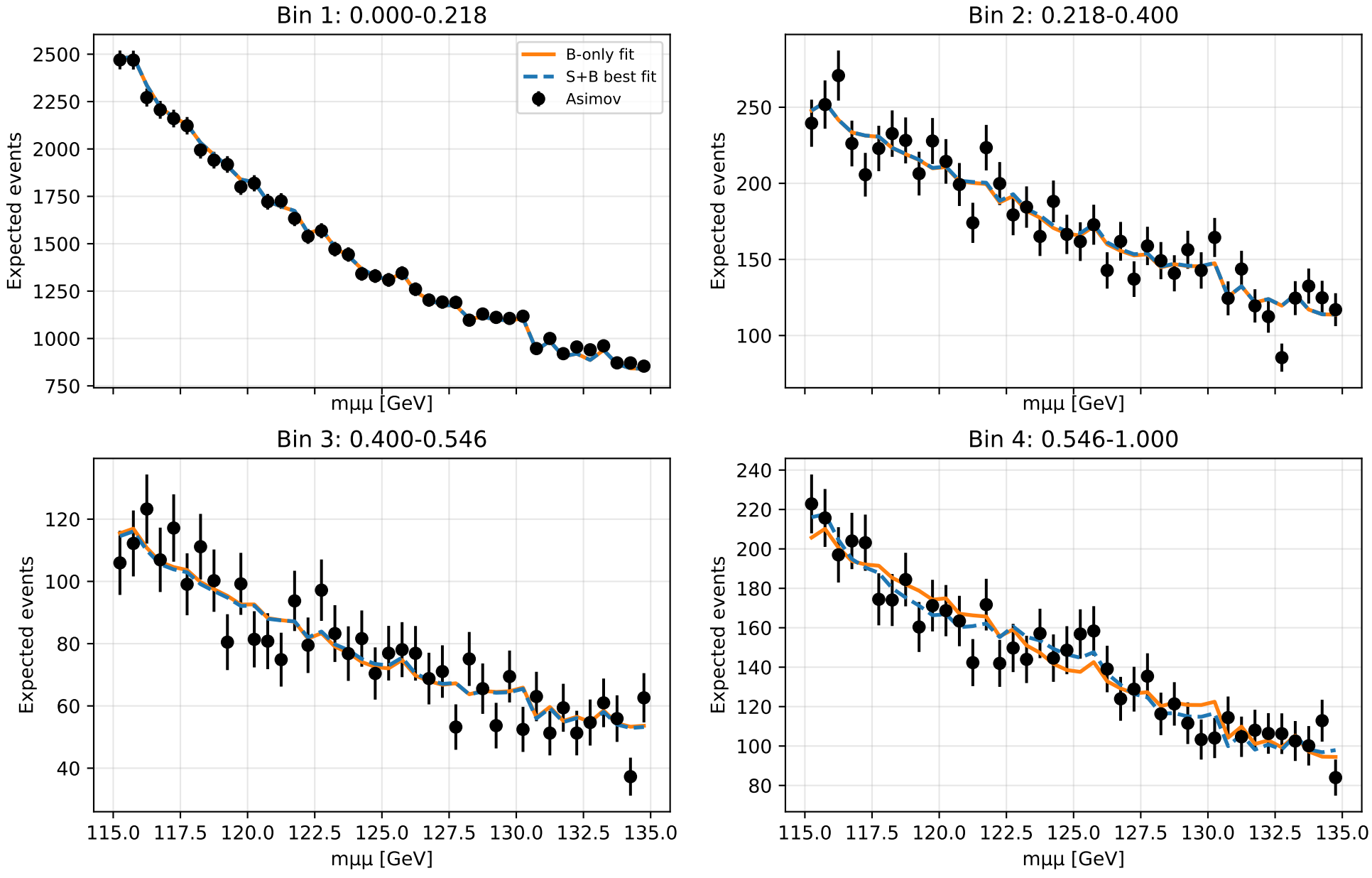
2024: Simultaneous shape-based profile likelihood summary

Quantity	Value
Expected Z	2.9993
q0	8.9961
mu_hat	9.4167
N_BDT bins	4
N_mass bins	40
poly order	2
bkg norm unc	0.200
poly unc	0.300
core poly order	2
core from low/med bins	True
DY >=2j unc	0.050
sig norm unc	0.050
sig mass shift unc [GeV]	0.200
sig resolution unc	0.100

2024: Shape-likelihood bin yields (Asimov)

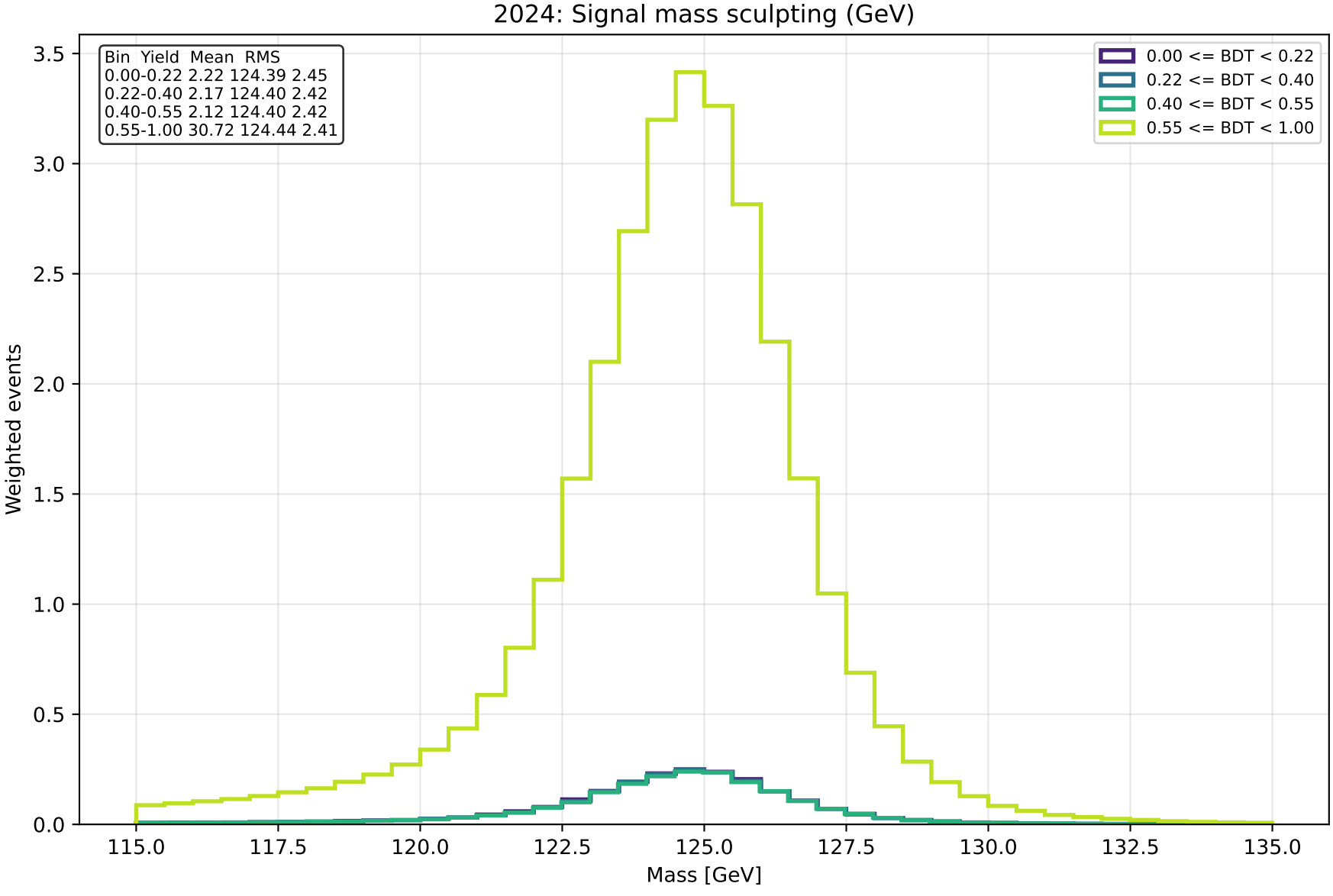
Bin	BDT	S truth	B truth	B-only fit	S+B fit
1	0.000-0.218	2.219	58322.830	58322.830	58343.748
2	0.218-0.400	2.168	6977.136	6977.136	6997.569
3	0.400-0.546	2.115	3110.638	3110.638	3105.793
4	0.546-1.000	30.723	5746.883	5793.042	5767.143

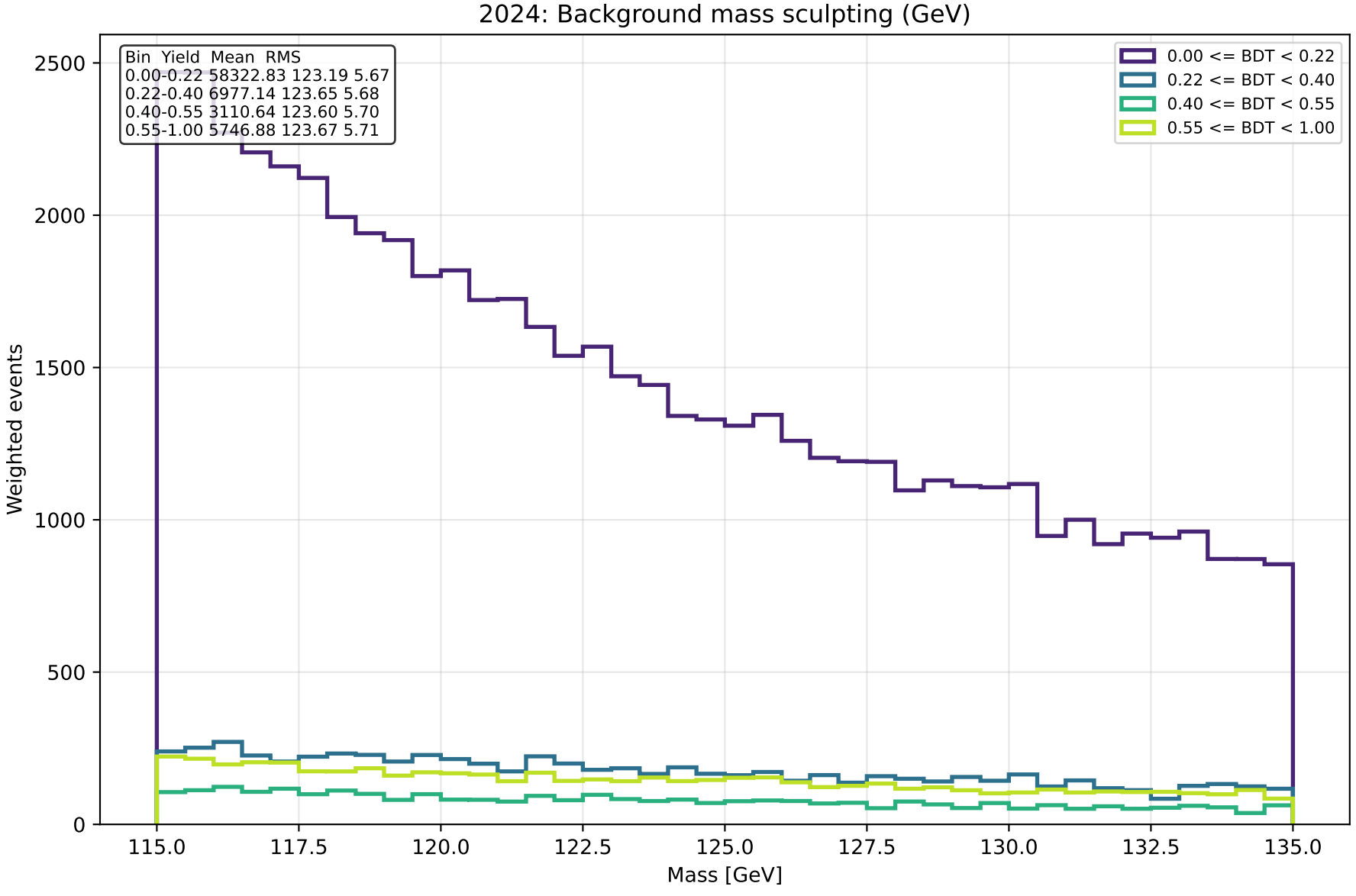

2024: Profile likelihood scan (simultaneous shape fit)
Zexp = 2.999
q0 = 8.996
μhat = 9.417
Δ(-2 ln L)
Signal strength μ
0
2
4
6
8
10

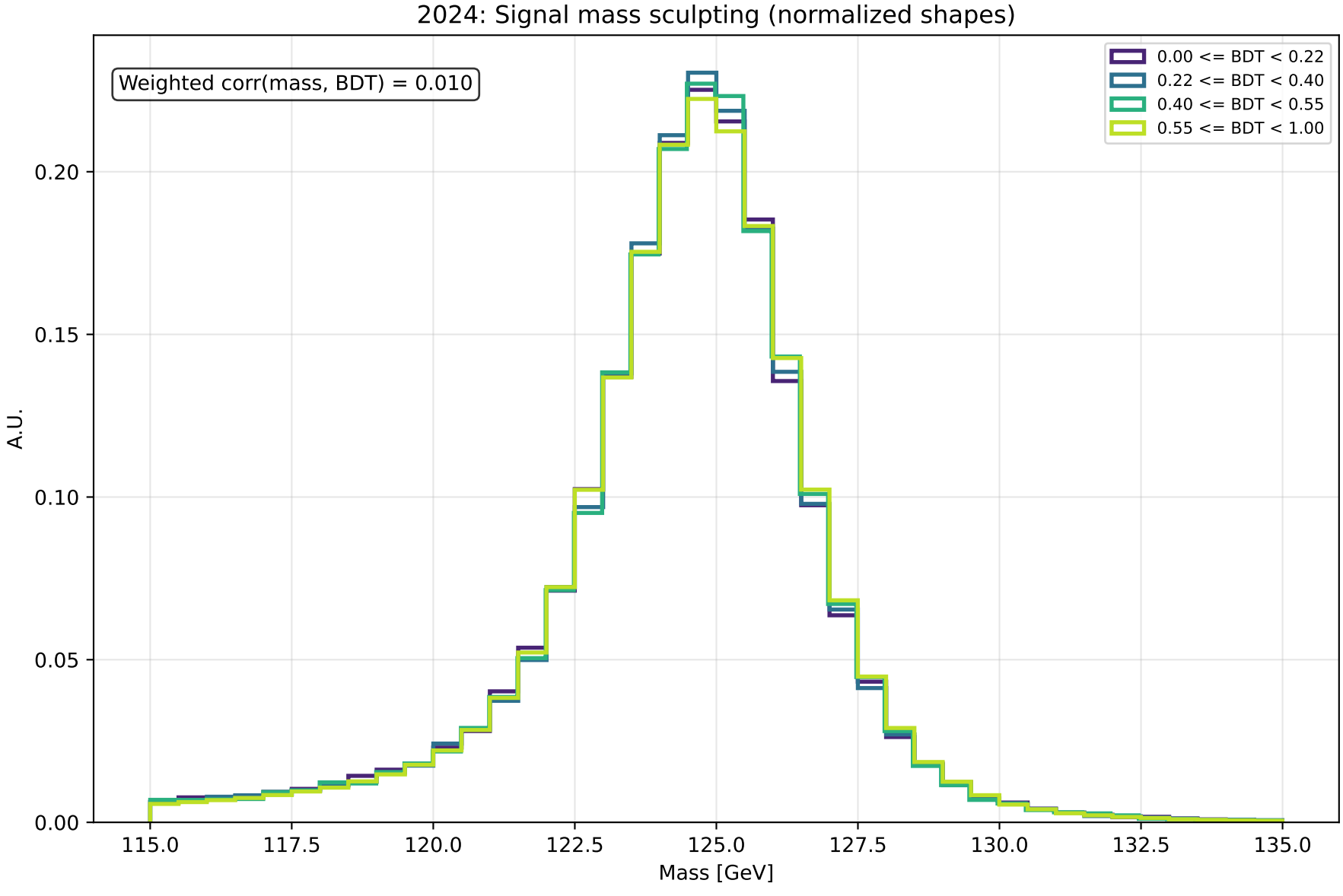

2024: Asimov mass templates by BDT bin (simultaneous fit)
Bin 1: 0.000-0.218
Bin 2: 0.218-0.400
Bin 3: 0.400-0.546
Bin 4: 0.546-1.000
B-only fit
S+B best fit
Asimov
Expected events
mμμ [GeV]

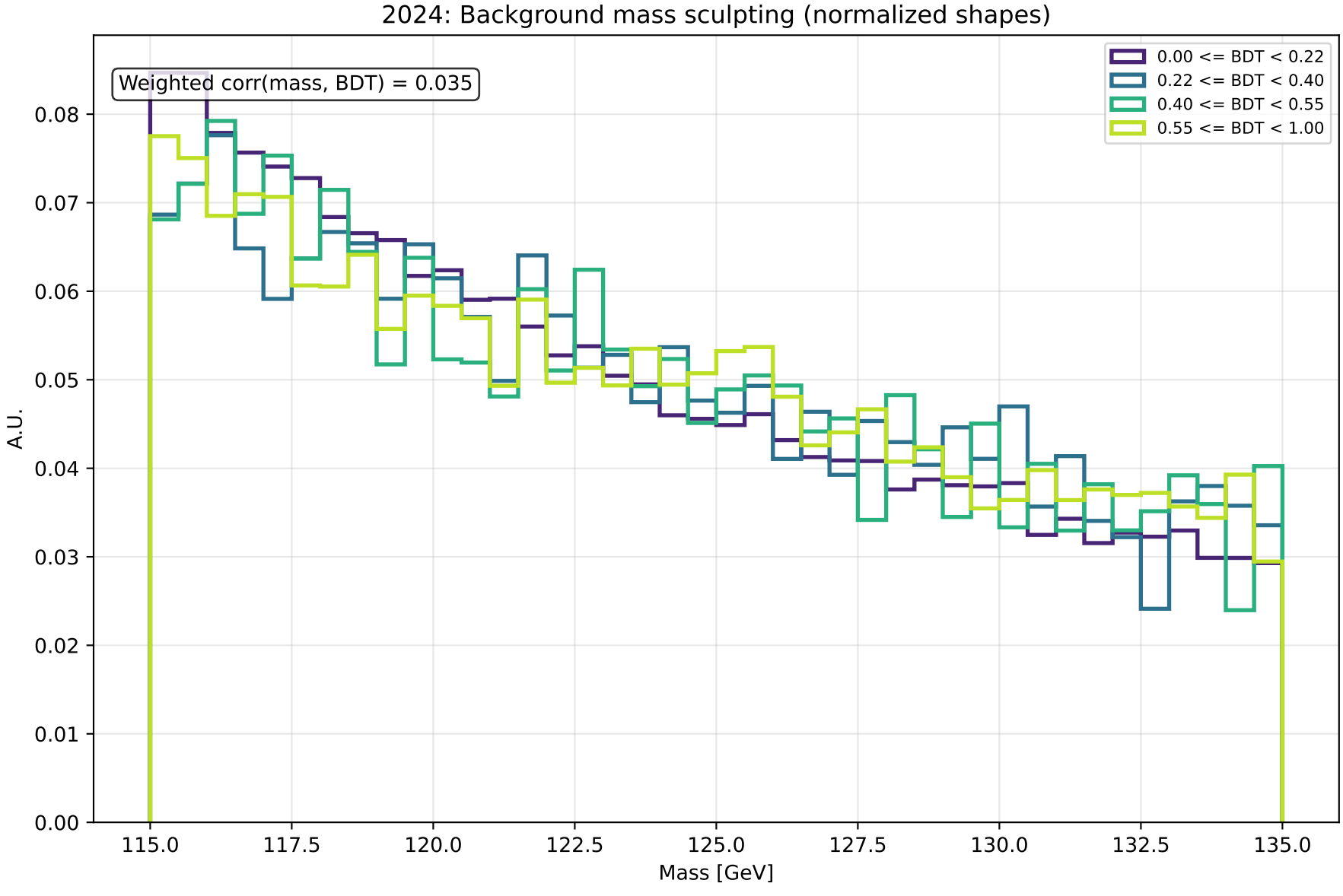
2024: Expected yield per optimized BDT bin (115-135 GeV)
S total=37.23, B total=74157.49

BDT bin	S (bin)	S frac	S>=lo	B (bin)	S/B (>=lo)
0.000-0.218	2.219	6.0%	100.0%	58322.830	0.000502
0.218-0.400	2.168	5.8%	94.0%	6977.136	0.00221
0.400-0.546	2.115	5.7%	88.2%	3110.638	0.00371
0.546-1.000	30.723	82.5%	82.5%	5746.883	0.00535

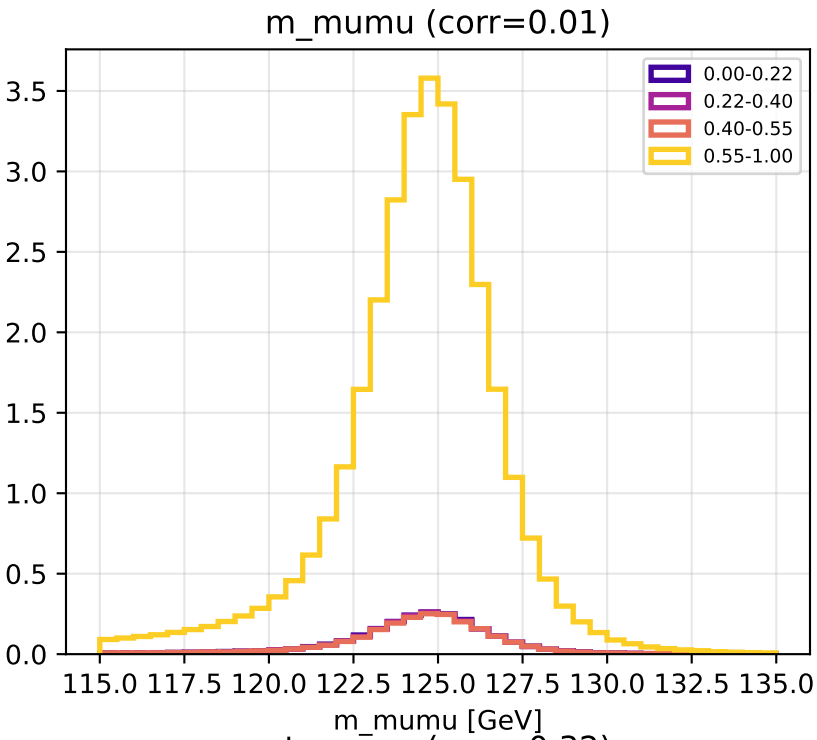

2024: Signal mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.22 2.22 124.39 2.45
0.22-0.40 2.17 124.40 2.42
0.40-0.55 2.12 124.40 2.42
0.55-1.00 30.72 124.44 2.41
0.00 <= BDT < 0.22
0.22 <= BDT < 0.40
0.40 <= BDT < 0.55
0.55 <= BDT < 1.00
Weighted events
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

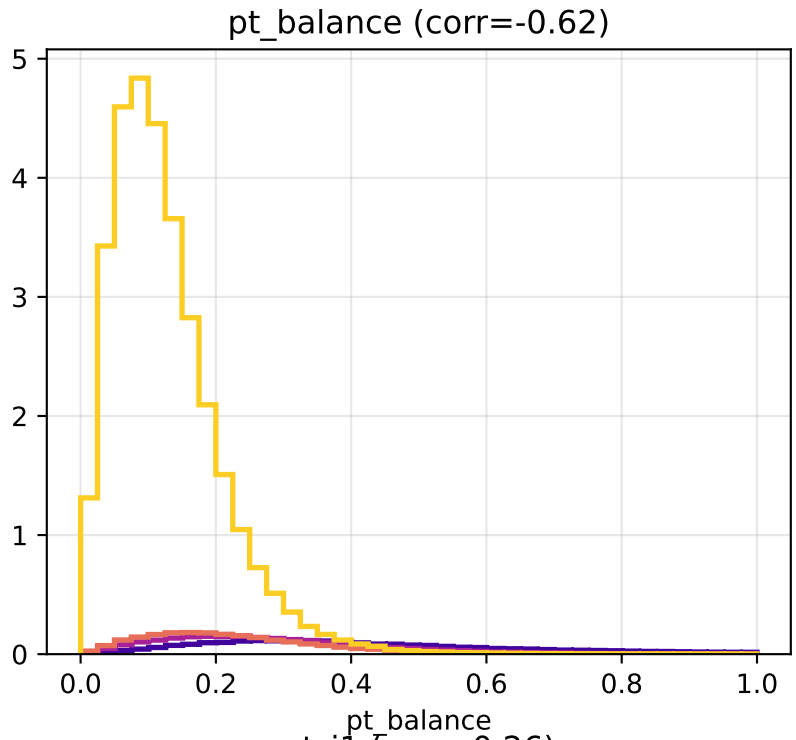

2024: Background mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.22 58322.83 123.19 5.67
0.22-0.40 6977.14 123.65 5.68
0.40-0.55 3110.64 123.60 5.70
0.55-1.00 5746.88 123.67 5.71
0.00 <= BDT < 0.22
0.22 <= BDT < 0.40
0.40 <= BDT < 0.55
0.55 <= BDT < 1.00
Weighted events
0
500
1000
1500
2000
2500
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

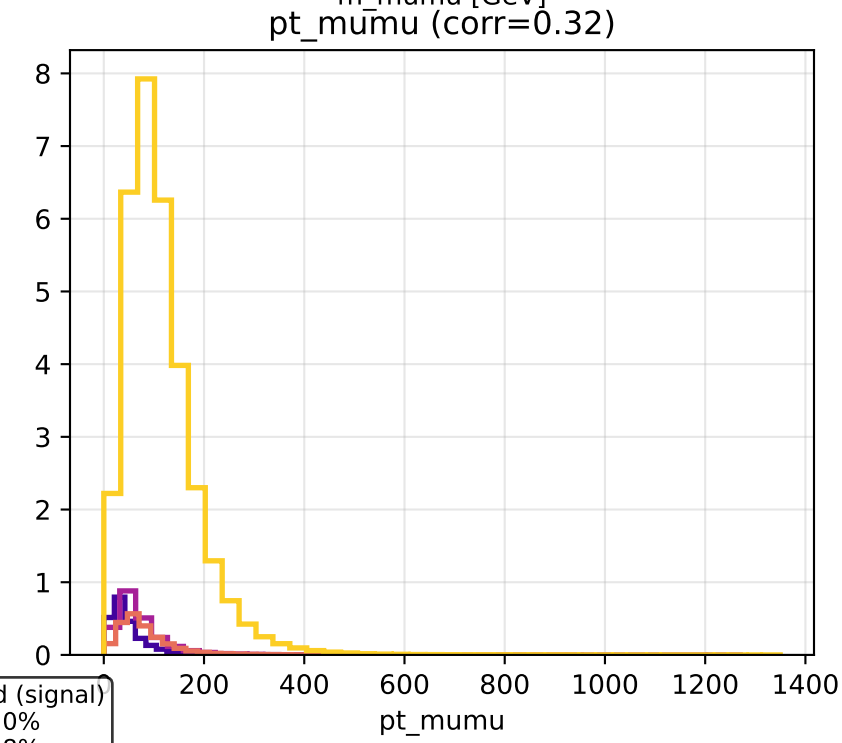

2024: Signal mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.010
0.00 <= BDT < 0.22
0.22 <= BDT < 0.40
0.40 <= BDT < 0.55
0.55 <= BDT < 1.00
A.U.
0.00
0.05
0.10
0.15
0.20
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

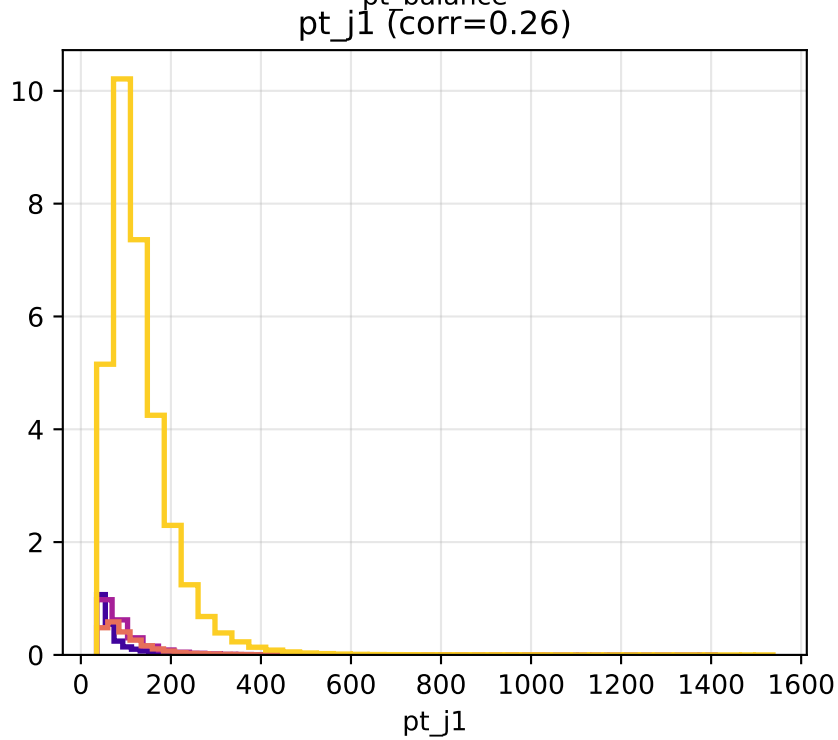

2024: Background mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.035
0.00 <= BDT < 0.22
0.22 <= BDT < 0.40
0.40 <= BDT < 0.55
0.55 <= BDT < 1.00
A.U.
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

2024: Signal distributions by BDT bin

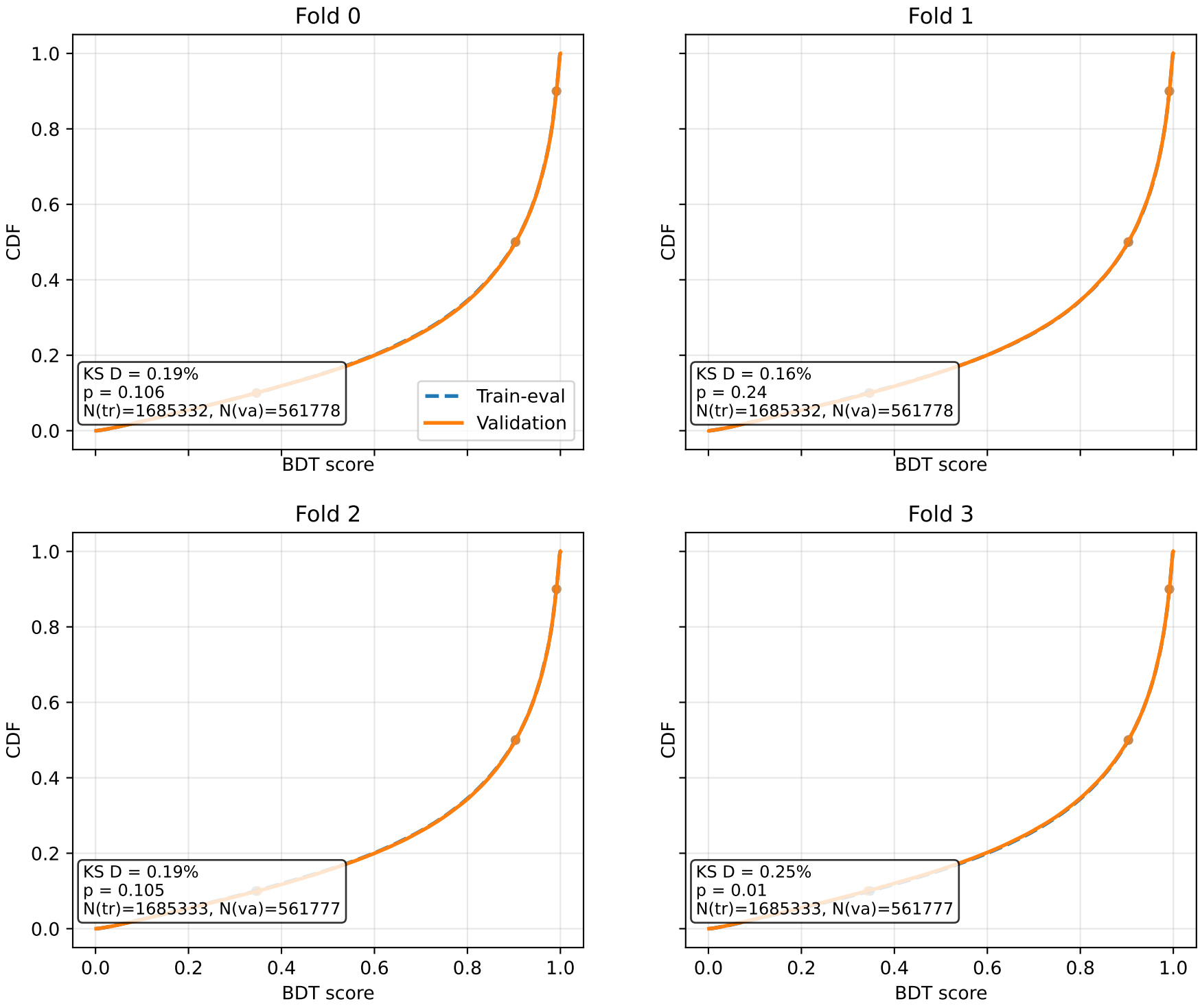








BDT bin yield (signal)
0.00-0.22: 6.0%
0.22-0.40: 5.8%
0.40-0.55: 5.7%
0.55-1.00: 82.5%


2024: CDF (train-eval vs validation)
Fold 0
KS D = 0.19%
p = 0.106
N(tr)=1685332, N(va)=561778
Train-eval
Validation
Fold 1
KS D = 0.16%
p = 0.24
N(tr)=1685332, N(va)=561778
Fold 2
KS D = 0.19%
p = 0.105
N(tr)=1685333, N(va)=561777
Fold 3
KS D = 0.25%
p = 0.01
N(tr)=1685333, N(va)=561777
CDF
BDT score
0.0
0.2
0.4
0.6
0.8
1.0

2024: KS test summary

Fold	KS statistic	N events	alpha	Pass threshold	KS pass
0	0.19%	561778	0.001	0.20%	TRUE
1	0.16%	561778	0.001	0.20%	TRUE
2	0.19%	561777	0.001	0.20%	TRUE
3	0.25%	561777	0.001	0.20%	FALSE

2024: Top-10 feature importance (gain)
2024_featureimportance_background.txt

Feature	Gain %
pt_balance	33.48%
pt_mumu	11.43%
pt_j1	11.10%
z	9.63%
dphi_mumu_jj	7.52%
deta_jj	7.02%
pt_j2	6.56%
njets	3.13%
jet1_eta	2.27%
mu1_pt_over_m	1.77%